

Assignment 5

1 Requirements

You are expected to complete all exercises. For submission, please put all your answers in a PDF file and submit it. The name of the file should follow the format "student-ID_assignmentNumber" (e.g., 2023111054_assignment5). **The submission deadline is 11:59 PM, Nov 20.**

2 Required Exercises (100 points)

Exercise 1: For the SDD in Figure 1, give annotated parse trees for the following expressions:

1. $(3 + 4) * (5 + 6)n$ [20points]
2. $1 * 2 * 3 * (4 + 5)n$ [20 points]
3. $(9 + 8 * (7 + 6) + 5) * 4n$ [20 points]

	PRODUCTION	SEMANTIC RULES
1)	$L \rightarrow E \mathbf{n}$	$L.val = E.val$
2)	$E \rightarrow E_1 + T$	$E.val = E_1.val + T.val$
3)	$E \rightarrow T$	$E.val = T.val$
4)	$T \rightarrow T_1 * F$	$T.val = T_1.val \times F.val$
5)	$T \rightarrow F$	$T.val = F.val$
6)	$F \rightarrow (E)$	$F.val = E.val$
7)	$F \rightarrow \mathbf{digit}$	$F.val = \mathbf{digit.lexval}$

Figure 1: Syntax-directed definition of a simple desk calculator

Exercise 2: What are all the topological sorts for the dependency graph of Figure 2? One sort mentioned during lecture is 1,2,3,...,9 (slide #17 of Chapter 4). [20 points]

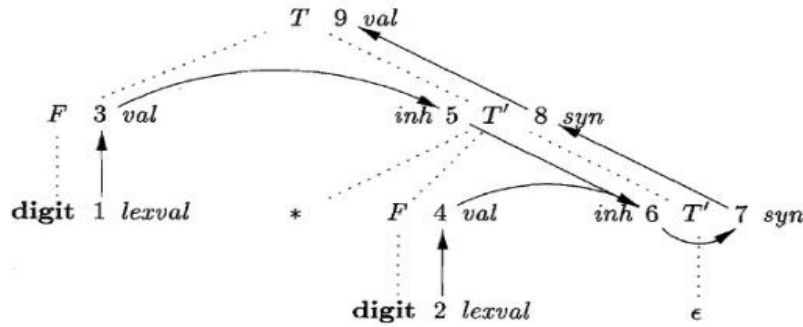


Figure 2: A dependency graph

Exercise 3: Below is a grammar for expressions involving operator $+$ and integer or floating-point operands. Floating-point numbers are distinguished by having a decimal point. Give an SDD to determine the type of each term T and expression E . [20 points]

$$E \rightarrow E + T \mid T$$

$$T \rightarrow num \cdot num \mid num$$